MATH 4665 Homework 4

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Question 1: Existence and Uniqueness of PDE Solutions

Consider the one-dimensional Poisson problem

$$-u''(x) = f(x), \qquad x \in (0,1), \qquad u(0) = u(1) = 0.$$

(a) Weak formulation in a Sobolev space

Let $v \in C_c^\infty(0,1)$. Multiplying the equation by v and integrating over $(0,1)$ gives

$$-\int_0^1 u''(x)v(x)\,dx = \int_0^1 f(x)v(x)\,dx.$$

An integration by parts yields

$$\int_0^1 u'(x)v'(x)\,dx = \int_0^1 f(x)v(x)\,dx,$$

because $v(0) = v(1) = 0$. This identity suggests the weak problem

$$\text{find } u \in V \text{ such that } a(u,v) = L(v) \quad \forall v \in V,$$

with

$$V = H_0^1(0,1), \qquad a(u,v) = \int_0^1 u'(x)v'(x)\,dx, \qquad L(v) = \int_0^1 f(x)v(x)\,dx.$$

The natural solution space is therefore $H_0^1(0,1)$, the closure of $C_c^\infty(0,1)$ in the H^1 norm. The test functions are initially taken in $C_c^\infty(0,1)$; by density, the weak identity extends to every $v \in H_0^1(0,1)$.

On $H_0^1(0,1)$ it is convenient to use the norm

$$\|u\|_V := \|u'\|_{L^2(0,1)}.$$

By the Poincaré inequality this norm is equivalent to the usual H^1 norm, so it defines the same topology on V. With this choice, the bilinear form is continuous:

$$|a(u,v)| \le \|u'\|_{L^2}\|v'\|_{L^2} = \|u\|_V\|v\|_V.$$

It is also coercive:

$$a(u,u) = \|u'\|_{L^2}^2 = \|u\|_V^2.$$

Thus a is continuous and coercive on V. If $f \in L^2(0,1)$, then L is a continuous linear functional on V since

$$|L(v)| \le \|f\|_{L^2}\|v\|_{L^2} \le C_P\|f\|_{L^2}\|v\|_V,$$

where C_P is the Poincaré constant. More generally, one may assume $f \in H^{-1}(0,1)$.

Example. If $f \equiv 1$, then the classical solution is

$$u(x) = \frac{x(1-x)}{2}.$$

Indeed, $u \in H_0^1(0,1)$, $-u'' = 1$, and for every $v \in H_0^1(0,1)$,

$$\int_0^1 u'(x)v'(x)\,dx = \int_0^1 v(x)\,dx,$$

so u satisfies the weak formulation.

(b) Lax–Milgram theorem and application

Lax–Milgram theorem. Let V be a Hilbert space, let $a : V \times V \to \mathbb{R}$ be continuous and coercive, and let $L \in V^*$. Then there exists a unique $u \in V$ such that

$$a(u,v) = L(v) \qquad \forall v \in V.$$

Moreover,

$$\|u\|_V \le \frac{1}{\alpha}\|L\|_{V^*},$$

where $\alpha > 0$ is the coercivity constant.

Proof idea. Let $J : V \to V^*$ be the Riesz isomorphism and define $\widetilde{A} := J^{-1}A : V \to V$, where $A : V \to V^*$ is given by

$$(Au)(v) = a(u,v).$$

Continuity of a implies that $\widetilde{A}$ is bounded. Coercivity gives

$$\alpha\|u\|_V^2 \le a(u,u) = (\widetilde{A}u, u)_V \le \|\widetilde{A}u\|_V\|u\|_V,$$

hence $\|\widetilde{A}u\|_V \ge \alpha\|u\|_V$. Therefore $\widetilde{A}$ is injective and its range is closed. To prove surjectivity, let $w \in V$ satisfy $w \perp R(\widetilde{A})$. Then

$$0 = (\widetilde{A}u, w)_V = a(u,w) \qquad \forall u \in V.$$

Choosing $u = w$ gives

$$0 = a(w,w) \ge \alpha\|w\|_V^2,$$

so $w = 0$. Hence $R(\widetilde{A})$ is dense and closed, therefore $R(\widetilde{A}) = V$. Thus $\widetilde{A}$ is bijective, which is equivalent to surjectivity of A. This proves existence, and injectivity gives uniqueness.

Application to the Poisson equation. For the present problem, $V = H_0^1(0,1)$ is a Hilbert space, $a(u,v) = \int_0^1 u'v'\,dx$ is continuous and coercive, and $L(v) = \int_0^1 fv\,dx$ is continuous whenever $f \in L^2(0,1)$. All assumptions of the Lax–Milgram theorem are therefore satisfied. Consequently,

there exists a unique weak solution $u \in H_0^1(0,1)$ such that

$$\int_0^1 u'(x)v'(x)\,dx = \int_0^1 f(x)v(x)\,dx \qquad \forall v \in H_0^1(0,1).$$

Question 2: Kernel Ridge Regression

Let $\Omega = [0,1]^d$. The KRR estimator is defined by

$$\hat{f}_{n,\lambda} \in \arg\min_{f \in \mathcal{H}_k} \left\{ \frac{1}{n}\sum_{i=1}^n (Y_i - f(X_i))^2 + \lambda \|f\|_{\mathcal{H}_k}^2 \right\}, \qquad \lambda > 0.$$

(a) Representer theorem and the finite-dimensional system

Set

$$V_n := \operatorname{span}\{k(\cdot, X_1), \dots, k(\cdot, X_n)\} \subset \mathcal{H}_k.$$

Every $f \in \mathcal{H}_k$ admits an orthogonal decomposition

$$f = f_\parallel + f_\perp, \qquad f_\parallel \in V_n, \quad f_\perp \perp V_n.$$

By the reproducing property,

$$f_\perp(X_i) = \langle f_\perp, k(\cdot, X_i)\rangle_{\mathcal{H}_k} = 0 \qquad \text{for } i = 1, \dots, n.$$

Hence the empirical loss depends only on $f_\parallel$, whereas

$$\|f\|_{\mathcal{H}_k}^2 = \|f_\parallel\|_{\mathcal{H}_k}^2 + \|f_\perp\|_{\mathcal{H}_k}^2.$$

If $f_\perp \neq 0$, the regularization term is strictly larger while the data-fit term is unchanged. Therefore every minimizer satisfies $f_\perp = 0$, and thus

$$\hat{f}_{n,\lambda}(x) = \sum_{i=1}^n \alpha_i k(x, X_i)$$

for some coefficient vector $\alpha = (\alpha_1, \dots, \alpha_n)^\top \in \mathbb{R}^n$.

Let $K \in \mathbb{R}^{n \times n}$ be the Gram matrix,

$$K_{ij} = k(X_i, X_j),$$

and let $Y = (Y_1, \dots, Y_n)^\top$. Then

$$\hat{f}_{n,\lambda}(X_j) = \sum_{i=1}^n \alpha_i k(X_j, X_i) = (K\alpha)_j,$$

and

$$\|\hat{f}_{n,\lambda}\|_{\mathcal{H}_k}^2 = \sum_{i,j=1}^n \alpha_i \alpha_j k(X_i, X_j) = \alpha^\top K \alpha.$$

The objective becomes

$$J(\alpha) = \frac{1}{n}\|Y - K\alpha\|_2^2 + \lambda \alpha^\top K \alpha.$$

Differentiating with respect to α gives

$$\nabla J(\alpha) = \frac{2}{n} K\big((K + n\lambda I)\alpha - Y\big).$$

Thus a canonical choice of coefficients is obtained from the linear system

$$(K + n\lambda I)\alpha = Y.$$

If K is singular, different coefficient vectors may represent the same function, but the fitted function $\hat{f}_{n,\lambda}$ is unique.

(b) Mean-square convergence and the role of λ_n

Let $T_k : L^2(\mu) \to L^2(\mu)$ be the kernel integral operator,

$$[T_k g](x) = \int_\Omega k(x,t)g(t)\, d\mu(t),$$

and let $(\mu_j, \phi_j)_{j\geq 1}$ be its eigenpairs. A standard consistency result for KRR assumes two ingredients.

First, the target function satisfies a source condition of the form

$$f_0 = T_k^r g \qquad \text{for some } r > 0 \text{ and } g \in L^2(\mu),$$

which quantifies the smoothness of f_0 relative to the kernel. Second, the spectrum of T_k is controlled through the effective dimension

$$\mathcal{N}(\lambda) := \mathrm{Tr}\big(T_k(T_k + \lambda I)^{-1}\big),$$

for example by an estimate

$$\mathcal{N}(\lambda) \leq C\lambda^{-\beta}, \qquad 0 < \beta \leq 1.$$

Under these assumptions, standard KRR theory gives the risk bound

$$R_n(\hat{f}_{n,\lambda}, f_0) = \mathbb{E}\|\hat{f}_{n,\lambda} - f_0\|^2_{L^2(\mu)} \leq C_1\lambda^{2r} + C_2\frac{\mathcal{N}(\lambda)}{n} \leq C_1\lambda^{2r} + C_3\frac{\lambda^{-\beta}}{n}.$$

The first term is the approximation error introduced by regularization; it vanishes when $\lambda \to 0$. The second term is the stochastic error produced by the noise and the random design; it decreases when λ is not too small.

Therefore, if one chooses a deterministic sequence λ_n such that

$$\lambda_n \to 0 \qquad \text{and} \qquad \frac{\mathcal{N}(\lambda_n)}{n} \to 0,$$

then both terms converge to zero and

$$R_n(\hat{f}_{n,\lambda_n}, f_0) \to 0.$$

Under the polynomial bound $\mathcal{N}(\lambda) \leq C\lambda^{-\beta}$, it is enough to require

$$\lambda_n \to 0, \qquad n\lambda_n^\beta \to \infty.$$

One common balancing choice is

$$\lambda_n \asymp n^{-1/(2r+\beta)},$$

which makes the bias and variance terms of the same order.

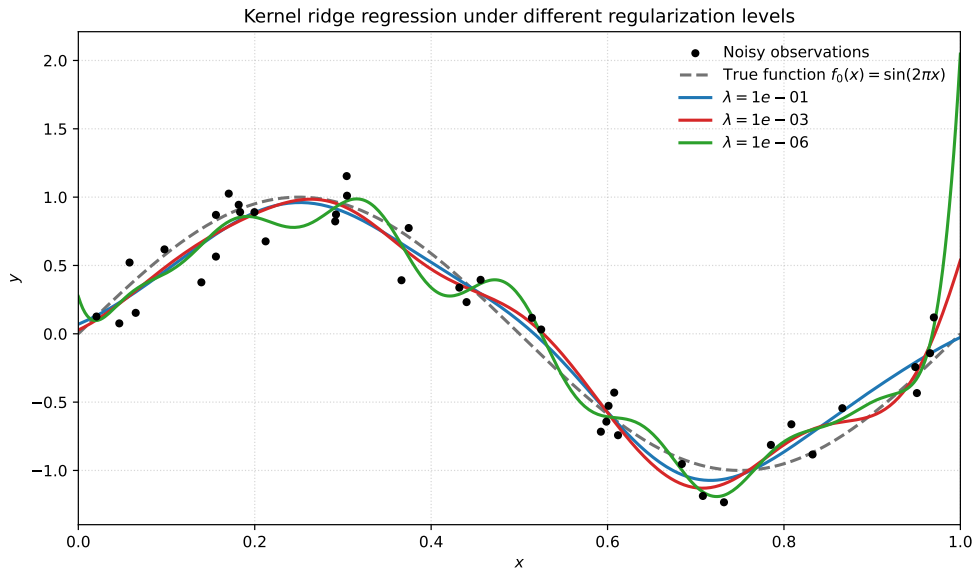

Figure 1: Kernel ridge regression on noisy one-dimensional data. Large λ oversmooths the estimator, while very small λ tracks the noise. An intermediate value gives the best compromise between bias and variance.

Question 3: Minimax Convergence Rate for KRR

(a) Meaning of the minimax risk

Let $\Theta \subseteq \mathcal{H}_k$ be a model class of admissible regression functions. The minimax risk is

$$R_n^* = \inf_{\tilde{f}_n} \sup_{f_0 \in \Theta} \mathbb{E}_{f_0} \|\tilde{f}_n - f_0\|^2_{L^2(\mu)}.$$

Parameter space. The set Θ specifies which true functions are allowed. In statistical terms, Nature chooses an unknown $f_0 \in \Theta$.

Loss function. The quantity being evaluated is the squared prediction error in $L^2(\mu)$,

$$\|\tilde{f}_n - f_0\|^2_{L^2(\mu)},$$

and the expectation is taken over the random sample and the observation noise.

Infimum over estimators. The statistician may choose any measurable estimator $\tilde{f}_n$ based on the data. Taking the infimum asks for the best possible procedure among all estimators.

Supremum over f_0. The supremum measures worst-case performance over the whole model class. An estimator is judged by the most difficult function in Θ, not by an easy or favorable case.

For this reason, R_n^* is a benchmark for the problem itself: no estimation method can have uniformly smaller risk than the minimax order over the entire class Θ.

(b) Bias–variance decomposition for KRR

For fixed design points $X_1, \ldots, X_n$, define the sampling operator

$$S_n : \mathcal{H}_k \to \mathbb{R}^n, \qquad S_n f = (f(X_1), \ldots, f(X_n))^\top,$$

and the empirical covariance operator

$$T_n := \frac{1}{n} S_n^* S_n.$$

Then the KRR estimator can be written as

$$\hat{f}_{n,\lambda} = (T_n + \lambda I)^{-1} \frac{1}{n} S_n^* Y.$$

Since $Y = S_n f_0 + \varepsilon$, where $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_n)^\top$, we have

$$\hat{f}_{n,\lambda} = (T_n + \lambda I)^{-1} T_n f_0 + (T_n + \lambda I)^{-1} \frac{1}{n} S_n^* \varepsilon.$$

Subtracting f_0 yields

$$\hat{f}_{n,\lambda} - f_0 = -\lambda (T_n + \lambda I)^{-1} f_0 + (T_n + \lambda I)^{-1} \frac{1}{n} S_n^* \varepsilon.$$

Taking conditional expectation with respect to the noise gives

$$\mathbb{E}_\varepsilon[\hat{f}_{n,\lambda} \mid X] = (T_n + \lambda I)^{-1} T_n f_0,$$

so the conditional mean-square error decomposes as

$$\begin{aligned} \mathbb{E}_\varepsilon\Big[\|\hat{f}_{n,\lambda} - f_0\|_{L^2(\mu)}^2 \mid X\Big] &= \Big\|\lambda (T_n + \lambda I)^{-1} f_0\Big\|_{L^2(\mu)}^2 \\ &\quad + \mathbb{E}_\varepsilon\left[\left\|(T_n + \lambda I)^{-1} \frac{1}{n} S_n^* \varepsilon\right\|_{L^2(\mu)}^2 \middle| X\right]. \end{aligned}$$

The cross term vanishes because $\mathbb{E}_\varepsilon[\varepsilon \mid X] = 0$. Averaging over the design produces the full risk

$$R_n(\hat{f}_{n,\lambda}, f_0) = \mathbb{E}_X[\mathrm{Bias}_n(\lambda, X)] + \mathbb{E}_X[\mathrm{Var}_n(\lambda, X)].$$

The random quantities are $X_1, \ldots, X_n$, $\varepsilon_1, \ldots, \varepsilon_n$, the operators S_n, T_n, and therefore the estimator $\hat{f}_{n,\lambda}$. The fixed quantities are the kernel k, the target function f_0, and the prescribed regularization level $\lambda = \lambda_n$.

The role of λ_n is to balance the two terms. A large λ_n strongly stabilizes the inverse $(T_n + \lambda_n I)^{-1}$, which suppresses variance but enlarges the bias. A small λ_n reduces bias but amplifies fluctuations caused by the noise. In the standard regime of Question 2 one obtains a proxy bound

$$R_n(\hat{f}_{n,\lambda}, f_0) \lesssim \lambda^{2r} + n^{-1} \lambda^{-\beta},$$

and balancing the two contributions gives

$$\lambda_n \asymp n^{-1/(2r+\beta)}.$$

This short computation explains why the regularization parameter determines the final convergence

rate.